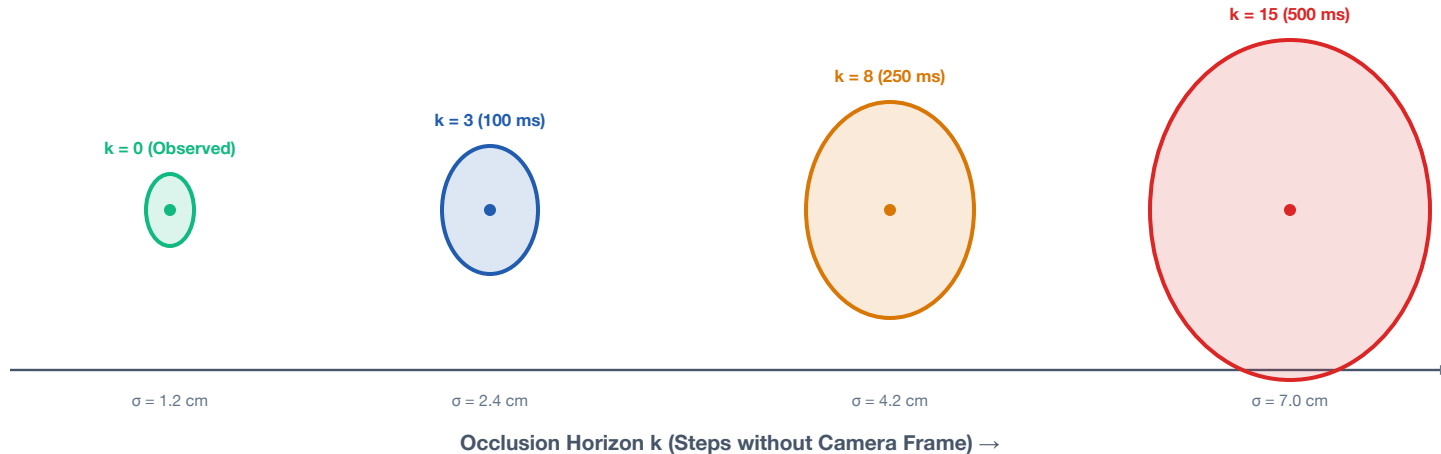


SPATIAL UNCERTAINTY EXPANSION UNDER SENSOR OCCLUSION

Open-Loop Covariance Growth: $\Sigma_{t+k} = F^k \Sigma_t (F^k)^T + \sum_{i=0}^{k-1} F^i Q (F^i)^T$



THE COVARIANCE EXPANSION INVARIANT

1. A world model must never treat dead-reckoned predictions as deterministic ground truth.
2. When covariance Σ_t exceeds the clearance margin ($3\sigma > d_{\text{clearance}}$), the CBF safety enforcer must trigger an immediate velocity back-off.